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OOPS Assignment 5

1. Try out all the code demonstrated in the lectures.

```
Dimensions are (1, 1, 1, 1.00)
Dimensions are (2, 4, 6)
Dimensions are (3, 5, 7, 2.00)
Dimensions are (4, 6, 8, 3.00)
Dimensions are (4, 6, 8, 3.00)
Dimensions are (4, 6, 8)
```

2. We have discussed the following example as part of lecture on Generalization.

Write a java program to implement the respective classes and have the inheritance among these classes.(4 points)

- a. In your test class demonstrate attributes of Employee being displayed by an object of PermanentEmployee
- b. Override the display method in the subclasses. (1 point)
- c. Epf is 25% of the salary. Implement calculateEPF for PermanentEmployee. (1 point)
- d. Use Super for creating the constructors. (1 point)
- e. Put daily wages for the TemporayEmployee and calculate his salary for the month of August 2026. (1 point)

```
Employee Name: tri Employee id: 100
```

```
Temporary Employee id : 102
Employee name: gru
Employee Salary: 3000.000000
Probationary period: 12
```

```
Permanent Employee id : 100
Employee name: tri
Employee Salary: 2000.000000
Employee epf amount 500.000000
```

```
Salary of August for gru is 200.0
```

3. You already have designed a BankAccount class as part of Exercise 3 – Question 7 (6 points)

- Have proper credit() withdraw() and displayBalance() methods.
 - Have a method to transfer some amount to a different BankAccount. (1 point)
- Create classes SavingsAccount and CheckingAccount as subclasses of BankAccount.
- Have interestRate as an attribute in SavingsAccount.
 - calculate the interest and add it to the balance of the account (1 point)
 - Checking account charges transaction fees.
 - Have some number of free transactions (e.g. 3) for the checking account as an attribute. This number should be a constant. (1 point)
 - Update the credit() withdraw() methods of CheckingAccount accordingly. Take advantage of the credit() and withdraw() implementations of the BankAccount. (2 points) Hint: use super
 - Have a method to deduct fees in CheckingAccount. (1 point)
 - In your test class demonstrate respective features to get the relevant points

```
Balance of Apple Banana's account is 12.0
Balance of Superman Batman's account is 10.0

Transferring 13Rs from a to b
Insufficient Bank balance
Transfer Failed!
Balance of Apple Banana's account is 12.0
Balance of Superman Batman's account is 10.0

Transferring 12Rs from a to b
12.0Rs withdrawn from Apple's account
12.0Rs credited to Superman's account
Balance of Apple Banana's account is 0.0
Balance of Superman Batman's account is 22.0

Self Transferring 9Rs from b to a
9.0Rs withdrawn from Superman's account
9.0Rs credited to Apple's account
Balance of Superman Batman's account is 13.0
Balance of Apple Banana's account is 9.0

Converting a into savings account with interest rate 25%
Balance of Apple Banana's savings account is 9.0 with interest rate 25.0
balance after adding interest on Apple's savings account: 11.25

Converting b into checking account
Balance of Superman Batman's checking account is 13.0 with no.of transactions done 0

3 sample transactions (3 is transaction limit)
1.0Rs withdrawn from Superman's account
3.0Rs credited to Superman's account
1.0Rs withdrawn from Superman's account
Balance of Superman Batman's checking account is 14.0 with no.of transactions done 3
Balance of Apple Banana's savings account is 11.25 with interest rate 25.0

Transferring 5Rs to savings account of Apple Banana
Transaction limit exceeded, proceed with extra fees (2Rs)? [yes = 1/no = 0]
1
5.0Rs withdrawn from Superman's account
5.0Rs credited to Apple's account
Balance of Apple Banana's savings account is 16.25 with interest rate 25.0
Balance of Superman Batman's checking account is 7.0 with no.of transactions done 4
```

```
Transferring 5Rs to savings account of Apple Banana
Transaction limit exceeded, proceed with extra fees (2Rs)? [yes = 1/no = 0]
0
Withdraw unsuccessfull
Transfer Failed!
Balance of Apple Banana's savings account is 11.25 with interest rate 25.0
Balance of Superman Batman's checking account is 14.0 with no.of transactions done 3
```